The University of Faisalabad

Public Key

What is Public Key Cryptography?

- Also known as asymmetric cryptography.
- Uses two keys: public key (shared) and private key (kept secret).
- One key encrypts, the other decrypts.
- Used in HTTPS, email encryption, and digital signatures.

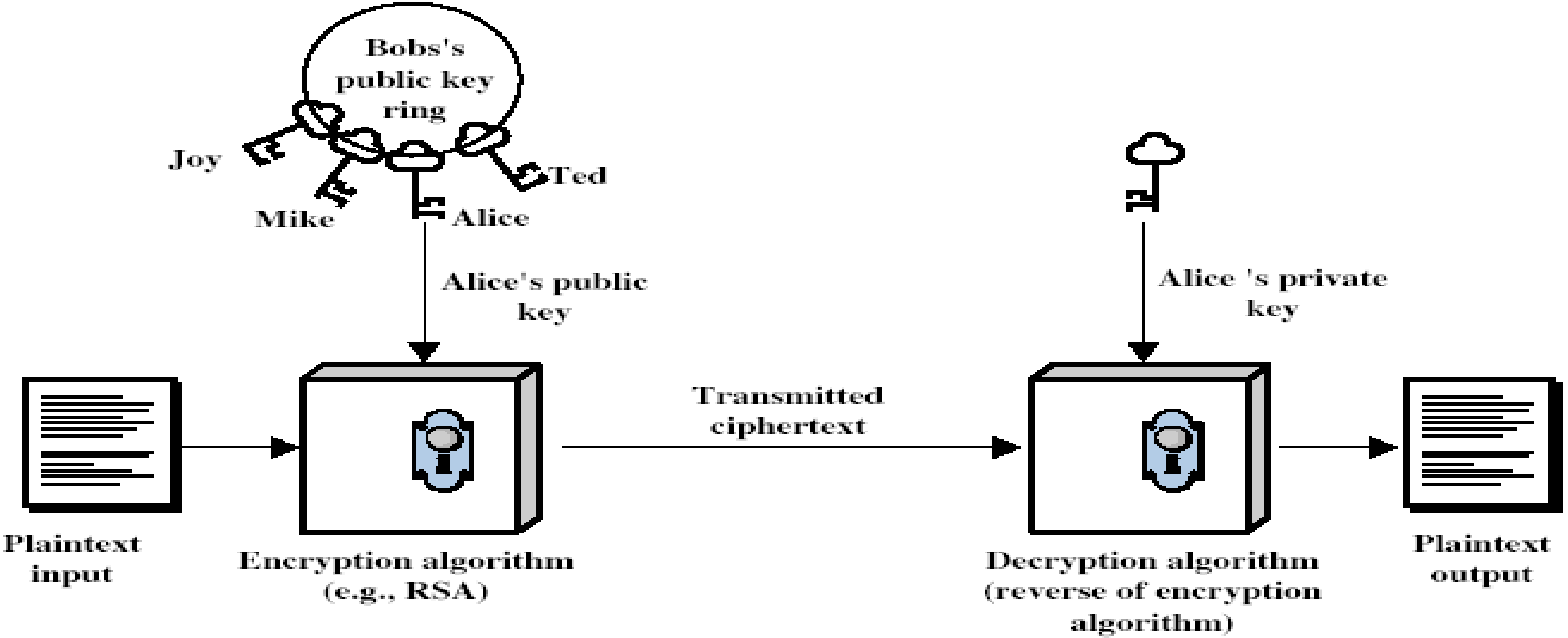
Public-Key Cryptography

- Probably most significant advance in the 3000 year history of cryptography
- Uses **two** keys – a public & a private key
- **Asymmetric** since parties are **not** equal
- Uses clever application of number theoretic concepts to function
- Complements **rather than** replaces private key crypto

Public-Key Cryptography

- **Public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
  - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
  - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- **Asymmetric** because
  - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

Public-Key Cryptography



Public-Key Characteristics

- Public-Key algorithms rely on two keys with the characteristics that it is:
  - computationally infeasible to find decryption key knowing only algorithm & encryption key
  - computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
  - either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)

RSA Algorithm Overview

- Most widely used public key algorithm.
- Based on the difficulty of factoring large primes.
- Key generation involves modulus (n), public exponent (e), and private exponent (d).
- Enables secure communication and authentication.

Given: C = 14, d = 27, n = 55

To Solve: 14^27 mod 55

Step 1: Use modular exponentiation:

14^1 mod 55 = 14

14^2 mod 55 = (14*14) mod 55 = 196 mod 55 = 31 14^4 mod 55 = (31*31) mod 55 = 961 mod 55 = 26 14^8 mod 55 = (26*26) mod 55 = 676 mod 55 = 16 14^16 mod 55 = (16*16) mod 55 = 256 mod 55 = 36

Example

Step 2: Multiply required powers to get 14^27

$14^{27} = 14^{16} \times 14^{8} \times 14^{2} \times 14^{1} \bmod 55$

$= (36 \times 16 \times 31 \times 14) \bmod 55$

Calculating step by step:

$36 \times 16 = 576 \bmod 55 = 26$

$26 \times 31 = 806 \bmod 55 = 36$

$36 \times 14 = 504 \bmod 55 = 9$

Final Answer: $14^{27} \bmod 55 = 9$

DES vs Public Key Cryptography

- DES is fast but uses the same key for encryption and decryption (symmetric).
- Public key cryptography is slower but more secure for key exchange.
- Both are used together in hybrid encryption systems.

Question?